# Mathematics II

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You can find the last version of these course materials at

https://mbujosab.github.io/MatematicasII/

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## Contents

**V Determinants** **1**

**LECTURE 14: Properties of Determinants** **2**
*Slides for Lecture 14* . . . . . 2
*Questions of the Lecture 14* . . . . . 8

**LECTURE 15: Cofactors and Determinant Formulas** **10**
*Slides for Lecture 15* . . . . . 10
*Questions of the Lecture 15* . . . . . 15

**Solutions** **19**

# Part V
# Determinants

# LECTURE 14: Properties of Determinants

## *Lecture 14*

(**Lecture** 14) S-1 Overview of Lesson 14

### Overview of *Lesson 14*

- Determinant: $\det(\mathbf{A}) \equiv |\mathbf{A}|$ $[\det : \mathbb{R}^{n\times n} \longrightarrow \mathbb{R}]$
  - Volume vs determinant
  - Properties: **1**, **2**, **3**
- We will derive the properties: **4** – **9**

F1

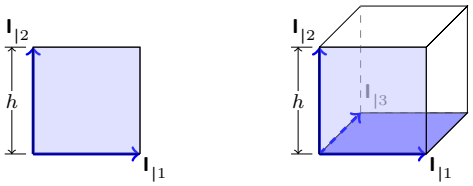
"Pinche aquí" y vea el notebook de Jupyter de la Lección 14. (¡Ojo! están mal numerados los Notebooks)

# Definition of the determinant function by three properties related to volume or area

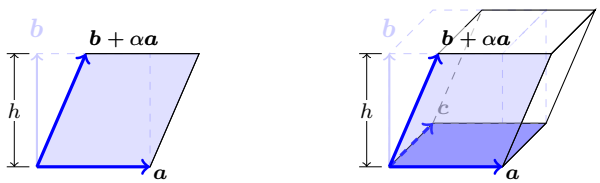
## Three properties of the area (volume) function of a parallelepiped

(**Lecture** 14) **S-2** **Area or Volume**

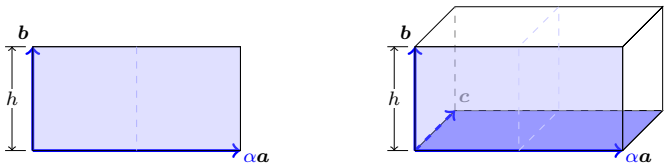
1. $\mathrm{Vol}\big(\underset{n\times n}{\mathbf{I}}\big) = 1.$




2. $\mathrm{Vol}(\mathbf{A}) = \mathrm{Vol}\Big(\mathbf{A}\underset{[(\alpha)\boldsymbol{k}+\boldsymbol{j}]}{\boldsymbol{\tau}}\Big)$ for $j \neq k$.




3. $|\alpha| \cdot \mathrm{Vol}(\mathbf{A}) = |\alpha| \cdot \mathrm{Vol}\big[\ldots;\mathbf{A}_{|k};\ldots\big] = \mathrm{Vol}\big[\ldots;\alpha\mathbf{A}_{|k};\ldots\big]$




F2

## The first three properties (P-1 to P-3)

By analogy with the volume function, we define the determinant as follows:

**Definition 1.** *The* determinant *is any function that assigns to each system of $n$ vectors in $\mathbb{R}^n$ (or to each square matrix of order $n$) a real number*

$$\det : \quad \mathbb{R}^{n\times n} \longrightarrow \mathbb{R}$$

*such that it satisfies the following three properties:*

(**Lecture** 14) **S-3** **Determinant: 3 properties that define the function**

**P-1** **Determinant of identity matrices:** **P-1**

$$\det \underset{n\times n}{\mathbf{I}} = 1$$

**P-2** ***Type I* elemen. transf. do not change the determinant**: **P-2**

$$\det \mathbf{A} = \det\left(\mathbf{A}_{\underset{[(\alpha)\boldsymbol{k}+\boldsymbol{j}]}{\boldsymbol{\tau}}}\right)$$

**P-3** **Multiplying a column by a scalar multiplies the determinant** **P-3**

$$\alpha \cdot \det \mathbf{A} \;=\; \det\left[\ldots;\alpha\mathbf{A}_{|k};\ldots\right] \text{ for any } k \in \{1:n\} \text{ and } \alpha \in \mathbb{R}$$

$$\boxed{\text{Absolute value of } \det \mathbf{A} = \operatorname{Vol} \mathbf{A}}$$

F3

Therefore, the absolute value of the determinant function is the volume function. We will use two alternative notations for the determinant of the matrix $\mathbf{A}$:

$$\text{determinant of } \mathbf{A} \equiv \det(\mathbf{A}) \equiv |\mathbf{A}|$$

**Advertencia:** Una barra vertical a cada lado de una matriz $|\mathbf{A}|$ significa el determinante de la matriz. Una barra vertical a cada lado de un número $|a|$ significa el valor absoluto del número. Es decir, el significado de las barras viene dado por el objeto encerrado: si es un número es el *valor absoluto*, y si es una matriz es el *determinante*. Jugando con esto, podemos decir que

$$\operatorname{Vol} \mathbf{A} = \text{Absolute value of } \det \mathbf{A} = \big|\det \mathbf{A}\big| = \big||\mathbf{A}|\big|.$$

*Example* 1. For $\mathbb{R}^3$:

$$\begin{vmatrix} a_1 & (b_1+\alpha c_1) & c_1 \\ a_2 & (b_2+\alpha c_2) & c_2 \\ a_3 & (b_3+\alpha c_3) & c_3 \end{vmatrix} = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix};$$

$$\det\left[\boldsymbol{a};\ (\boldsymbol{b}+\alpha\boldsymbol{c});\ \boldsymbol{c};\right] = \det\left[\boldsymbol{a};\ \boldsymbol{b};\ \boldsymbol{c};\right];$$

and also

$$\begin{vmatrix} a_1 & \alpha\, b_1 & c_1 \\ a_2 & \alpha\, b_2 & c_2 \\ a_3 & \alpha\, b_3 & c_3 \end{vmatrix} = \alpha \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix};$$

$$\det\left[\boldsymbol{a};\ \alpha\boldsymbol{b};\ \boldsymbol{c};\right] = \alpha \det\left[\boldsymbol{a};\ \boldsymbol{b};\ \boldsymbol{c};\right];$$

Well, these three properties define the determinant function... now let us deduce everything else

## Additional properties (P-4 to P-11)

## Determinant of a matrix with a zero column

(**Lecture** 14) S-4 Determinant of a matrix with a zero column

**P-4** **Det. of a matrix $\mathbf{A}$ with a zero column** **P-4**

*If* $\mathbf{A}$ *has a zero column* $\mathbf{0}$*, then*

$$\boxed{\det(\mathbf{A}) = 0}$$

F5

To solve in classroom

Exercise 1. Let $\mathbf{A}$ be a square matrix of order $n$ with a zero column $\mathbf{0}$. Prove that its determinant is zero:

## Elementary transformations by columns.

(**Lecture** 14) S-5 Elementary matrices

We already know

$$\det\left(\mathbf{A}_{\underset{[(\alpha)k+j]}{\tau}}\right) = |\mathbf{A}|; \qquad \det\left(\mathbf{A}_{\underset{[(\alpha)k]}{\tau}}\right) = \alpha|\mathbf{A}|.$$

**Determinant of elementary matrices**

$$\det\left(\mathbf{I}_{\underset{[(\alpha)k+j]}{\tau}}\right) = 1 \qquad \text{and} \qquad \det\left(\mathbf{I}_{\underset{[(\alpha)j]}{\tau}}\right) = \alpha.$$

Hence, since $\mathbf{A}_{\tau} = \mathbf{A}(\mathbf{I}_{\tau})$, then

$$\left|\mathbf{A}(\mathbf{I}_{\tau})\right| = |\mathbf{A}| \cdot |\mathbf{I}_{\tau}| \tag{1}$$

where $\mathbf{I}_{\tau}$ is an elementary matrix

F6

## Sequence of elementary transformations by columns.

Exercise 2. Prove the following propositions

(a) $\det(\mathbf{A}_{\tau_1 \cdots \tau_k}) = |\mathbf{A}| \cdot |\mathbf{I}_{\tau_1}| \cdots |\mathbf{I}_{\tau_k}|$ .

(b) If $\mathbf{B}$ is a full rank matrix, i.e., if $\mathbf{B} = \mathbf{I}_{\tau_1 \cdots \tau_k}$, then $|\mathbf{B}| = |\mathbf{I}_{\tau_1}| \cdots |\mathbf{I}_{\tau_k}|$, and therefore $|\mathbf{B}| \neq 0$.

(c) *If* $\mathbf{A}$ *and* $\mathbf{B}$ *have order* $n$ *and* $\mathbf{B}$ *is full rank, then*

$$\det(\mathbf{AB}) = |\mathbf{A}| \cdot |\mathbf{B}| \tag{2}$$

(**Lecture** 14) S-6 Determinant after a sequence of elementary transformations

*Example* 2. A sequence $\tau_1 \cdots \tau_k$ of *Type I* elementary transformations of the matrix $\mathbf{A}$ does not change the determinant.

$$|\mathbf{A}_{\tau_1 \cdots \tau_k}| = |\mathbf{A}(\mathbf{I}_{\tau_1 \cdots \tau_k})| = |\mathbf{A}| \cdot |\mathbf{I}_{\tau_1 \cdots \tau_k}| = |\mathbf{A}| \cdot 1 = |\mathbf{A}|$$

*Example* 3. But a sequence of *Type II* transformations can do so.

$$\begin{vmatrix} 2a & 3c \\ 2b & 3d \end{vmatrix} = \underline{?} \begin{vmatrix} a & c \\ b & d \end{vmatrix}$$

F7

## Antisymmetric property. Determinant of a singular matrix, the inverse, and the product of matrices

### Column permutation

(**Lecture** 14) S-7 Antisymmetric property

**P-5** **[Antisymmetric property]** **P-5**

*Exchanging two columns changes the sign of the determinant.*

*Proof.* A column exchange is a sequence of *Type I* transformations and a single *Type II* transformation that multiplies one column by $-1$ □

Therefore:

$$\begin{vmatrix} c_1 & b_1 & a_1 \\ c_2 & b_2 & a_2 \\ c_3 & b_3 & a_3 \end{vmatrix} = (-1) \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$$

F8

### Singular matrices. Inverse of a matrix

**Note 1.** *By elementary transformations, it is possible to reduce* $\underset{n\times n}{\mathbf{A}}$ *to its reduced echelon form* $\mathbf{R}$.

*There are two cases: if the matrix is singular (rank $< n$), then the determinant is zero; if the matrix is full rank,* $\mathbf{R} = \mathbf{I}$*; and we only need to consider the* Type II *elementary operations we used to get* $\mathbf{I}$ *(the* Type I *ones do not matter!...)*

(**Lecture** 14) S-8 Singular matrices. Inverse of a matrix

**P-6** **If $\mathbf{A}$ is singular then $|\mathbf{A}| = 0$.** **P-6**

**P-7** $\det(\mathbf{A}^{-1}) = \left(\det \mathbf{A}\right)^{-1}$**.** **P-7**

*Proof.* Let $\mathbf{A}_{\tau_1 \cdots \tau_k} = \underset{n\times n}{\mathbf{R}}$ be a reduced echelon form (and $\mathbf{E} = \mathbf{I}_{\tau_1 \cdots \tau_k}$).

Since $\mathbf{AE} = \mathbf{R}$, then: $|\mathbf{A}| \cdot |\mathbf{E}| = |\mathbf{R}|$; with only two cases:

$$\begin{cases} \mathbf{A} \text{ singular } (\mathbf{R}_{|n} = \mathbf{0}): & |\mathbf{A}| \cdot |\mathbf{E}| = 0 \;\Rightarrow\; |\mathbf{A}| = 0 \\ \mathbf{A} \text{ not singular } (\mathbf{R} = \mathbf{I}): & |\mathbf{A}| \cdot |\mathbf{E}| = 1 \;\Rightarrow\; |\mathbf{E}| = \left|\mathbf{A}^{-1}\right| = \left(|\mathbf{A}|\right)^{-1} \end{cases}.$$

□

F9

### Determinant of the inverse of a matrix

Moreover, we can calculate the determinant of $\mathbf{A}^{-1}$ by Gaussian elimination. If $\mathbf{R} = \mathbf{I}$ then $|\mathbf{A}| \cdot |\mathbf{E}| = |\mathbf{A}| \cdot |\mathbf{A}^{-1}| = |\mathbf{I}| = 1$ and therefore

$$\boxed{|\mathbf{A}^{-1}| = \left(|\mathbf{A}|\right)^{-1}.}$$

*Example* 4. For $\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 2 & 2 \end{bmatrix}$ :

$$\begin{array}{c} \begin{bmatrix} 1 & 2 \\ 2 & 2 \end{bmatrix} \\ \hline \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \end{array} \xrightarrow[Type I]{\overset{\tau}{[(-2)\mathbf{1}+\mathbf{2}]}} \begin{array}{c} \begin{bmatrix} 1 & 0 \\ 2 & -2 \end{bmatrix} \\ \hline \begin{bmatrix} 1 & -2 \\ 0 & 1 \end{bmatrix} \end{array} \xrightarrow[Type II]{\overset{\tau}{[(-1/2)\mathbf{2}]}} \begin{array}{c} \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} \\ \hline \begin{bmatrix} 1 & 1 \\ 0 & -1/2 \end{bmatrix} \end{array} \xrightarrow[Type I]{\overset{\tau}{[(-2)\mathbf{2}+\mathbf{1}]}} \begin{array}{c} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ \hline \begin{bmatrix} -1 & 1 \\ 1 & -1/2 \end{bmatrix} \end{array}$$

Thus

$$\left|\mathbf{A}^{-1}\right| = \left|\mathbf{I}_{\overset{\tau}{[(-2)\mathbf{1}+\mathbf{2}]}}\right| \cdot \left|\mathbf{I}_{\overset{\tau}{[(-1/2)\mathbf{2}]}}\right| \cdot \left|\mathbf{I}_{\overset{\tau}{[(-2)\mathbf{2}+\mathbf{1}]}}\right| = 1 \cdot \frac{-1}{2} \cdot 1 = \frac{-1}{2};$$

that is $\boxed{|\mathbf{A}| = -2.}$

**Determinant of the product.**

Remember that for any $\mathbf{B}$ of order $n$, there exists $\mathbf{E} = \mathbf{I}_{\tau_1 \cdots \tau_k}$ (full rank) such that $\mathbf{BE} = \mathbf{L}$ is an echelon form of $\mathbf{B}$. If $\mathbf{B}$ is singular, then $\mathbf{L}_{|n} = \mathbf{0}$. Thus, applying the same elementary column transformations to $\mathbf{AB}$, that is, calculating the product $\mathbf{ABE}$, gives us

$$\mathbf{ABE}_{|n} = \mathbf{AL}_{|n} = \mathbf{0};$$

since $\mathbf{E}_{|n} \neq \mathbf{0}$, it follows that $(\mathbf{AB})$ is singular .

(**Lecture** 14) S-9 **Determinant of a product of matrices**

P-8 **[Determinant of a product of matrices]** P-8

$$\boxed{\det(\mathbf{AB}) = \det(\mathbf{A}) \cdot \det(\mathbf{B}).} \tag{3}$$

$$\begin{cases} \mathbf{B} \text{ singular, then so it is } \mathbf{AB} \Rightarrow & \det(\mathbf{AB}) = 0 = \det(\mathbf{A}) \cdot \det(\mathbf{B}) \\ \mathbf{B} = \mathbf{I}_{\tau_1 \cdots \tau_k} \Rightarrow & \det(\mathbf{AB}) = \det(\mathbf{A}) \cdot \det(\mathbf{B}) \end{cases}$$

F10

## Determinant of the Transposed Matrix.

To solve in classroom

Exercise 3. **[Transposed matrices]**

(a) What is the relationship between the determinant of an elementary matrix $\mathbf{I}_\tau$ and the determinant of its transpose ${}_\tau\mathbf{I}$?

(b) Let $\mathbf{B}$ be a full rank matrix, prove that $|\mathbf{B}| = |\mathbf{B}^\mathsf{T}|$.

(**Lecture** 14) S-10 Determinant of a Transpose

**P-9** **Determinant of a Transpose** **P-9**

$$|\mathbf{A}| = |\mathbf{A}^{\mathsf{T}}|.$$

*Proof.*

$$\begin{cases} \text{if } \mathbf{A} \text{ es singular:} & \mathbf{A}^{\mathsf{T}} \text{ es singular } \Rightarrow \det \mathbf{A}^{\mathsf{T}} = \det \mathbf{A} = 0 \\ \text{if } \mathbf{A} \text{ NO es singular:} & \mathbf{A} = \mathbf{I}_{\tau_1 \cdots \tau_k} \Rightarrow \det \mathbf{A}^{\mathsf{T}} = \det \mathbf{A} \end{cases}.$$

□

F11

The lecture ends here

## *Questions of the Lecture 14*

(L-14) Question 1. Complete the proofs of this lecture.

(L-14) Question 2. Knowing that the determinant of the product of any two matrices $\mathbf{B}$ and $\mathbf{C}$ is $|\mathbf{BC}| = |\mathbf{B}||\mathbf{C}|$; prove that for any invertible matrix $\mathbf{A}$ (therefore, with $\det \mathbf{A} \neq 0$)

$$\det(\mathbf{A}^{-1}) = \Big(\det(\mathbf{A})\Big)^{-1}.$$

(L-14) Question 3. Consider $\underset{3\times 3}{\mathbf{A}}$ and $\underset{3\times 3}{\mathbf{B}}$ such that $\det(\mathbf{A}) = 2$ and $\det(\mathbf{B}) = -2$

(a) ($0.5^{\text{pts}}$) Compute the determinants of $\mathbf{A}(\mathbf{B})^2$ and $(\mathbf{AB})^{-1}$
(b) ($0.5^{\text{pts}}$) Is it possible to compute the rank of $\mathbf{A} + \mathbf{B}$? and of $\mathbf{AB}$?

(L-14) Question 4. Use the Gauss-Jordan method to compute the determinant of the following matrices

(a) $\mathbf{A}_1 = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$

(b) $\mathbf{A}_2 = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$

(c) $\mathbf{A}_3 = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$

(L-14) Question 5. The 3 by 3 matrix $\mathbf{A}$ reduces to the identity matrix $\mathbf{I}$ through the following three column elementary operations (in order):

$\underset{[(-4)\mathbf{1}+\mathbf{2}]}{\tau}$: Subtract 4 times column 1 from column 2.

$\underset{[(-3)\mathbf{1}+\mathbf{3}]}{\tau}$: Subtract 3 times column 1 from column 3.

$\underset{[(-1)\mathbf{3}+\mathbf{2}]}{\tau}$: Subtract column 3 from column 2.

Calculate the determinant of $\mathbf{A}$.

(L-14) Question 6.

(a) Find the determinant of $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & c \\ 0 & 0 & 1 \end{bmatrix}$ and $\begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$

(b) Find the determinant of $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ a & b & c & d \end{bmatrix}$ using Gauss-Jordan.

***End of Questions of the Lecture 14***

# LECTURE 15: Cofactors and Determinant Formulas

## *Lecture 15*

(**Lecture** 15) S-1 Overview of Lesson 15

### Overview of *Lesson 15*

- Computing $|\mathbf{A}|$ by Gaussian elimination
- **P-10** — Multilinear property
- Expansion of det $\mathbf{A}$ in cofactors (Laplace expansion).
- Applications of the determinant function
  - Cramer's rule for solving a system of equations
  - Computing the inverse of $\mathbf{A}$

F12

"Pinche aquí" y vea el cuaderno de Jupyter de la Lección 15

(**Lecture** 15) S-2 Extended matrix

$$\text{Extended matrix of } \mathbf{B}: \quad \begin{bmatrix} \mathbf{B} & \\ & 1 \end{bmatrix}$$

1. Given $\boldsymbol{\tau}$: $\begin{bmatrix} \mathbf{B}_{\boldsymbol{\tau}} & \\ & 1 \end{bmatrix} = \begin{bmatrix} \mathbf{B} & \\ & 1 \end{bmatrix}_{\boldsymbol{\tau}}$.
2. Since $\begin{bmatrix} \mathbf{I} & \\ & 1 \end{bmatrix}_{\boldsymbol{\tau}}$ and $\mathbf{I}_{\boldsymbol{\tau}}$ same *type* Elem. Mat. $\Rightarrow$ same det.

Applying 1. $k$ times, and then 2.

$$\left|\begin{bmatrix} \mathbf{I}_{\boldsymbol{\tau}_1\cdots\boldsymbol{\tau}_k} & \\ & 1 \end{bmatrix}\right| = \left|\begin{bmatrix} \mathbf{I} & \\ & 1 \end{bmatrix}_{\boldsymbol{\tau}_1\cdots\boldsymbol{\tau}_k}\right| = \left|\begin{bmatrix} \mathbf{I} & \\ & 1 \end{bmatrix}_{\boldsymbol{\tau}_1}\cdots\begin{bmatrix} \mathbf{I} & \\ & 1 \end{bmatrix}_{\boldsymbol{\tau}_k}\right| = \left|\mathbf{I}_{\boldsymbol{\tau}_1}\right|\cdots\left|\mathbf{I}_{\boldsymbol{\tau}_k}\right| = \left|\mathbf{I}_{\boldsymbol{\tau}_1\cdots\boldsymbol{\tau}_k}\right|.$$

If $\mathbf{A}$ is the extended matrix of $\mathbf{B}$ $\begin{cases} \text{If } \mathbf{B} \text{ es singular} & |\mathbf{B}| = 0 = |\mathbf{A}| \\ \text{If } \mathbf{B} \text{ es invertible} & |\mathbf{B}| = |\mathbf{A}| \end{cases}$

F13

## Triangular matrices

Exercise 4. **[Triangular matrices]**
(a) Find the determinant of a full rank lower triangular matrix $\mathbf{L}$
(b) What is the determinant of a triangular matrix with a zero entry in its diagonal?
(c) What is the determinant of an upper triangular matrix? $\mathbf{U}$

Additionally $\begin{vmatrix} \mathbf{A} & \underset{m\times n}{\mathbf{0}} \\ \underset{n\times m}{\mathbf{C}} & \mathbf{B} \end{vmatrix} = |\mathbf{A}| \cdot |\mathbf{B}|$.

## Cálculo del determinante por eliminación gaussiana

(**Lecture** 15) $\boxed{\text{S-3}}$ Computation by Gaussian elimination

*Example* 5. $\mathbf{A} = \begin{bmatrix} 1 & 5 \\ 2 & 3 \end{bmatrix}$ : $\left[\begin{array}{cc|c} 1 & 5 & \\ 2 & 3 & \\ \hline & & 1 \end{array}\right] \xrightarrow{\overset{\boldsymbol{\tau}}{[(-5)\mathbf{1}+\mathbf{2}]}} \left[\begin{array}{cc|c} 1 & 0 & \\ 2 & -7 & \\ \hline & & 1 \end{array}\right]$ $\boxed{|\mathbf{A}| = -7}$

*Example* 6.

$$\left[\begin{array}{ccc|c} 0 & 2 & 1 & 0 \\ 9 & 6 & 3 & 0 \\ 0 & 1 & 1 & 0 \\ \hline 0 & 0 & 0 & 1 \end{array}\right] \xrightarrow[\overset{\boldsymbol{\tau}}{[(\frac{1}{2})\mathbf{4}]}]{\overset{\boldsymbol{\tau}}{[(2)\mathbf{3}]}\ [(-1)\mathbf{2}+\mathbf{3}]} \left[\begin{array}{ccc|c} 0 & 2 & 0 & 0 \\ 9 & 6 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ \hline 0 & 0 & 0 & \frac{1}{2} \end{array}\right] \xrightarrow[\overset{\boldsymbol{\tau}}{[(-1)\mathbf{4}]}]{\overset{\boldsymbol{\tau}}{[\mathbf{1} \rightleftharpoons \mathbf{2}]}} \left[\begin{array}{ccc|c} 2 & 0 & 0 & 0 \\ 6 & 9 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ \hline 0 & 0 & 0 & -\frac{1}{2} \end{array}\right]$$

$$\begin{vmatrix} 0 & 2 & 1 \\ 9 & 6 & 3 \\ 0 & 1 & 1 \end{vmatrix} = -9,$$

F14

```
A =  Matrix([ [0,2,1], [9,6,3], [0,1,1] ])
Determinante(A,1)
```

## Simple formulas for matrices of order less than 4

### Matrices of order 1, $\mathbf{A} = [a]$ :

$$\left[\begin{array}{c|c} a & 0 \\ \hline 0 & 1 \end{array}\right] \Rightarrow |\mathbf{A}| = a.$$

### Matrices of order 2:

$$\left[\begin{array}{cc|c} a & b & 0 \\ c & d & 0 \\ \hline 0 & 0 & 1 \end{array}\right] \xrightarrow{\overset{\boldsymbol{\tau}}{[(-\frac{b}{a})\mathbf{1}+\mathbf{2}]}} \left[\begin{array}{cc|c} a & 0 & 0 \\ c & d-\frac{bc}{a} & 0 \\ \hline 0 & 0 & 1 \end{array}\right]$$

$$|\mathbf{A}| = ad - bc = a\det[d] - b\det[c].$$

### Matrices of order 3:

$$\left[\begin{array}{ccc|c} a & b & c & 0 \\ d & e & f & 0 \\ g & h & i & 0 \\ \hline 0 & 0 & 0 & 1 \end{array}\right] \xrightarrow[\overset{\boldsymbol{\tau}}{[(-\frac{c}{a})\mathbf{1}+\mathbf{3}]}]{\overset{\boldsymbol{\tau}}{[(-\frac{b}{a})\mathbf{1}+\mathbf{2}]}} \left[\begin{array}{ccc|c} a & 0 & 0 & 0 \\ d & e-\frac{bd}{a} & f-\frac{cd}{a} & 0 \\ g & h-\frac{bg}{a} & i-\frac{cg}{a} & 0 \\ \hline 0 & 0 & 0 & 1 \end{array}\right] \xrightarrow{\overset{\boldsymbol{\tau}}{[(\frac{-af+cd}{ae-bd})\mathbf{2}+\mathbf{3}]}} \left[\begin{array}{ccc|c} a & 0 & 0 & 0 \\ d & e-\frac{bd}{a} & 0 & 0 \\ g & h-\frac{bg}{a} & \frac{aei-afh-bdi+bfg+cdh-ceg}{ae-bd} & 0 \\ \hline 0 & 0 & 0 & 1 \end{array}\right]$$

$$|\mathbf{A}| = \underbrace{aei - afh - bdi + bfg + cdh - ceg}_{\text{(Rule of Sarrus)}} = a\begin{vmatrix} e & f \\ h & i \end{vmatrix} - b\begin{vmatrix} d & f \\ g & i \end{vmatrix} + c\begin{vmatrix} d & e \\ g & h \end{vmatrix}.$$

### Matrices of order 4:

$$\left[\begin{array}{cccc|c} a & b & c & d & 0 \\ e & f & g & h & 0 \\ i & j & k & l & 0 \\ m & n & o & p & 0 \\ \hline 0 & 0 & 0 & 0 & 1 \end{array}\right] \xrightarrow[\substack{\overset{\boldsymbol{\tau}}{[(-\frac{b}{a})\mathbf{1}+\mathbf{2}]} \\ [(-\frac{c}{a})\mathbf{1}+\mathbf{3}] \\ [(-\frac{d}{a})\mathbf{1}+\mathbf{4}] \\ [(\frac{-ag+ce}{af-be})\mathbf{2}+\mathbf{3}] \\ [(\frac{-ah+de}{af-be})\mathbf{2}+\mathbf{4}] \\ [(\frac{-afl+ahj+bel-bhi-dej+dfi}{afk-agj-bek+bgi+cej-cfi})\mathbf{3}+\mathbf{4}]}}$$

$$\left[\begin{array}{cccc|c} a & 0 & 0 & 0 & 0 \\ e & f-\frac{be}{a} & 0 & 0 & 0 \\ i & j-\frac{bi}{a} & \frac{afk-agj-bek+bgi+cej-cfi}{af-be} & 0 & 0 \\ m & n-\frac{bm}{a} & \frac{afo-agn-beo+bgm+cen-cfm}{af-be} & p+\frac{\left(-l+\frac{\left(h-\frac{de}{a}\right)\left(j-\frac{bi}{a}\right)}{f-\frac{be}{a}}+\frac{di}{a}\right)\left(-o+\frac{\left(g-\frac{ce}{a}\right)\left(n-\frac{bm}{a}\right)}{f-\frac{be}{a}}+\frac{cm}{a}\right)}{-k+\frac{\left(g-\frac{ce}{a}\right)\left(j-\frac{bi}{a}\right)}{f-\frac{be}{a}}+\frac{ci}{a}} - \frac{\left(h-\frac{de}{a}\right)\left(n-\frac{bm}{a}\right)}{f-\frac{be}{a}} - \frac{dm}{a} & 0 \\ \hline 0 & 0 & 0 & 0 & 1 \end{array}\right]$$

$$|\mathbf{A}| = afkp - aflo - agjp + agln + ahjo - ahkn - bekp + belo + bgip - bglm - bhio + bhkm + cejp - celn - cfip + cflm + chin - chjm - dejo + dekn + dfio - dfkm - dgin + dgjm = a\begin{vmatrix} f & g & h \\ j & k & l \\ n & o & p \end{vmatrix} - b\begin{vmatrix} e & g & h \\ i & k & l \\ m & o & p \end{vmatrix} + c\begin{vmatrix} e & f & h \\ i & j & l \\ m & n & p \end{vmatrix} - d\begin{vmatrix} e & f & g \\ i & j & k \\ m & n & o \end{vmatrix}$$

# Expansion of the determinant by cofactors (Laplace expansion).

## Multilinear property

(**Lecture** 15) S-4 Multilinear property

**P-10** **Multilinear property** **P-10**

$$\det\left[\ldots;(\beta \boldsymbol{b} + \psi \boldsymbol{c});\ldots\right] = \beta \det\left[\ldots;\boldsymbol{b};\ldots\right] + \psi \det\left[\ldots;\boldsymbol{c};\ldots\right]$$

*Example* 7. Then, in the 2-dimensional case $\mathbb{R}^2$

$$\begin{vmatrix} a+\alpha & c \\ b+\beta & d \end{vmatrix} = \begin{vmatrix} a & c \\ b & d \end{vmatrix} + \begin{vmatrix} \alpha & c \\ \beta & d \end{vmatrix};$$

that is

$$\begin{vmatrix} a & c \\ b & d \end{vmatrix} = \begin{vmatrix} a & c \\ 0 & d \end{vmatrix} + \begin{vmatrix} & c \\ & d \end{vmatrix}.$$

F17

*Proof.* In the book (`https://mbujosab.github.io/CursoDeAlgebraLineal/libro.pdf`) □

## Minors and cofactors

### New notation and definition of minors and cofactors

Consider $\boldsymbol{q} = (q_1, \ldots q_n,)$ in $\mathbb{R}^n$, if we remove the $j$th element, $q_j$, we will denote the new vector in $\mathbb{R}^{(n-1)}$ as

$$\boldsymbol{q}^{\not{j}} = \left(q_1, \ldots,\ q_{j-1},\quad q_{j+1},\ \ldots, q_n\right) \in \mathbb{R}^{n-1}.$$

In the same way, if $\mathbf{A}$ is of order $m$ by $n$, we denote the submatrix that results from removing the $i$-th row with ${}^{\not{i}}\mathbf{A}$, and the submatrix that results from removing the $j$-th column with $\mathbf{A}^{\not{j}}$. Therefore, ${}^{\not{i}}\mathbf{A}^{\not{j}}$, is the $m-1$ by $n-1$ submatrix that results from removing the $i$-th row and the $j$-th column.

(**Lecture** 15) S-5 minors and cofactors

**Definition 2** (minors and cofactors)**.** *We denote a submatrix of* $\mathbf{A}$ *obtained by deleting row* $i$ *and column* $j$ *of* $\mathbf{A}$ *by*

$${}^{\not{i}}\mathbf{A}^{\not{j}};$$

*Its determinant is called the* minor *of* $a_{ij}$*. Additionally,*

$$\operatorname{cof}_{ij}\left(\mathbf{A}\right) = (-1)^{i+j} \det\left({}^{\not{i}}\mathbf{A}^{\not{j}}\right)$$

*is called the* cofactor *of* $a_{ij}$*.*

F18

*Example* 8. For $\mathbf{A} = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$, we have

$$ {}^{1\urcorner}\mathbf{A}^{\urcorner 2} = \begin{bmatrix} 4 & 6 \\ 7 & 9 \end{bmatrix}, \qquad {}^{3\urcorner}\mathbf{A}^{\urcorner 3} = \begin{bmatrix} 1 & 2 \\ 4 & 5 \end{bmatrix} $$

hence

$$ \operatorname{cof}_{12}(\mathbf{A}) = (-1)^{1+2} \det\left({}^{1\urcorner}\mathbf{A}^{\urcorner 2}\right) = (-1) \begin{vmatrix} 4 & 6 \\ 7 & 9 \end{vmatrix}. $$

and

$$ \operatorname{cof}_{33}(\mathbf{A}) = (-1)^{3+3} \det\left({}^{3\urcorner}\mathbf{A}^{\urcorner 3}\right) = \begin{vmatrix} 1 & 2 \\ 4 & 5 \end{vmatrix}. $$

## Expansion by cofactors (Laplace expansion).

(**Lecture** 15) S-6 **Expansion by cofactors**

**Theorem 0.1** ([**Laplace expansion**]). *For any $n$ by $n$ matrix $\mathbf{A}$, $\det(\mathbf{A})$ can be expressed as the sum of the products of the elements of any column (row) of $\mathbf{A}$ by their corresponding cofactors:*

$$ \det(\mathbf{A}) = \sum_{i=1}^{n} a_{ij} \operatorname{cof}_{ij}(\mathbf{A}), \qquad \textit{the expansion by the } j\textit{th column} $$

*or*

$$ \det(\mathbf{A}) = \sum_{j=1}^{n} a_{ij} \operatorname{cof}_{ij}(\mathbf{A}), \qquad \textit{the expansion by the } i\textit{th row} $$

F20

*Proof.* In the book (`https://mbujosab.github.io/CursoDeAlgebraLineal/libro.pdf`) □

To solve in classroom

Exercise 5. Compute $\det \mathbf{A} = \begin{vmatrix} 2 & 0 & 3 & 2 \\ 5 & 1 & 2 & 4 \\ 3 & 0 & 1 & 2 \\ 5 & 3 & 2 & 1 \end{vmatrix}$

# First applications

## Cramer's Rule for solving a system of equations.

(**Lecture** 15) $\boxed{\text{S-7}}$ **Cramer's Rule**

$$\mathbf{A}\boldsymbol{x} = \boldsymbol{b}; \qquad |\mathbf{A}| \neq 0 \quad \text{then}$$

$$\boldsymbol{b} = (\mathbf{A}_{|1})x_1 + \cdots + (\mathbf{A}_{|j})x_j + \cdots + (\mathbf{A}_{|n})x_n.$$

$$\det\Big[\mathbf{A}_{|1};\ \ldots\ \overbrace{\boldsymbol{b}}^{\text{pos. } j};\ \ldots\ \mathbf{A}_{|n}\Big] = x_j \cdot \det(\mathbf{A}).$$

$$x_j = \frac{\det\Big[\mathbf{A}_{|1};\ \ldots\ \overbrace{\boldsymbol{b}}^{\text{pos. } j};\ \ldots\ \mathbf{A}_{|n}\Big]}{\det(\mathbf{A})}.$$

Computational issues when $\det \mathbf{A} \simeq 0$ (tiny angle between vectors) $\boxed{\text{F21}}$

# Second application

## Computing the inverse of a matrix

**Definition 3.** *For a* $\underset{n\times n}{\mathbf{A}}$ *matrix, the matrix* $\mathbf{Adj}(\mathbf{A})$ *(the* adjoint matrix *of* $\mathbf{A}$*) is defined as the transpose of the matrix obtained by replacing each element* $a_{ij}$ *of* $\mathbf{A}$ *with its corresponding cofactor* $\mathrm{cof}_{ij}(\mathbf{A})$*. That is,*

$$\mathbf{Adj}(\mathbf{A}) = \begin{bmatrix} \mathrm{cof}_{11}(\mathbf{A}) & \mathrm{cof}_{12}(\mathbf{A}) & \cdots & \mathrm{cof}_{1n}(\mathbf{A}) \\ \mathrm{cof}_{21}(\mathbf{A}) & \mathrm{cof}_{22}(\mathbf{A}) & \cdots & \mathrm{cof}_{2n}(\mathbf{A}) \\ \vdots & \vdots & \ddots & \vdots \\ \mathrm{cof}_{n1}(\mathbf{A}) & \mathrm{cof}_{n2}(\mathbf{A}) & \cdots & \mathrm{cof}_{nn}(\mathbf{A}) \end{bmatrix}^{\mathsf{T}}$$

What do we obtain if we multiply the adjoint matrix of $\mathbf{A}$ by $\mathbf{A}$?

(**Lecture** 15) S-8 **Computing the inverse of a matrix**

$$[\mathbf{Adj}(\mathbf{A})] \cdot \mathbf{A} = \begin{bmatrix} \mathrm{cof}_{11}(\mathbf{A}) & \mathrm{cof}_{21}(\mathbf{A}) & \cdots & \mathrm{cof}_{n1}(\mathbf{A}) \\ \mathrm{cof}_{12}(\mathbf{A}) & \mathrm{cof}_{22}(\mathbf{A}) & \cdots & \mathrm{cof}_{n2}(\mathbf{A}) \\ \vdots & \vdots & \ddots & \vdots \\ \mathrm{cof}_{1n}(\mathbf{A}) & \mathrm{cof}_{2n}(\mathbf{A}) & \cdots & \mathrm{cof}_{nn}(\mathbf{A}) \end{bmatrix} \underbrace{\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix}}_{\mathbf{A}}$$

F22

The first element on the diagonal of the resulting matrix is the expansion of $|\mathbf{A}|$ along the first column of $\mathbf{A}$. The second element on the diagonal is the expansion of $|\mathbf{A}|$ by the second column of $\mathbf{A}$, etc.

The elements outside the diagonal are determinants of matrices with two equal columns. For example, the second element of the first row of $[\mathbf{Adj}(\mathbf{A})] \cdot \mathbf{A}$ is the expansion along the first column of

$$\begin{vmatrix} a_{12} & a_{12} & \dots & a_{1n} \\ a_{22} & a_{22} & \dots & a_{2n} \\ a_{32} & a_{32} & \dots & a_{3n} \\ \vdots & \vdots & & \vdots \\ a_{n2} & a_{n2} & \dots & a_{nn} \end{vmatrix} = a_{12}\,\mathrm{cof}_{11}(\mathbf{A}) + a_{22}\,\mathrm{cof}_{21}(\mathbf{A}) + a_{32}\,\mathrm{cof}_{31}(\mathbf{A}) + \cdots + a_{n2}\,\mathrm{cof}_{n1}(\mathbf{A}) = \sum_{i=1}^{n} a_{i2}\,\mathrm{cof}_{i1}(\mathbf{A}) = 0,$$

where we find that the second column appears twice (in the first and second positions), and the determinant is therefore equal to zero.

And the $k$-th entry ($k \neq 1$) from the first row is

$$\begin{vmatrix} a_{1k} & a_{12} & \dots & a_{1n} \\ a_{2k} & a_{22} & \dots & a_{2n} \\ a_{3k} & a_{32} & \dots & a_{3n} \\ \vdots & \vdots & & \vdots \\ a_{nk} & a_{n2} & \dots & a_{nn} \end{vmatrix} = a_{1k}\,\mathrm{cof}_{11}(\mathbf{A}) + a_{2k}\,\mathrm{cof}_{21}(\mathbf{A}) + a_{3k}\,\mathrm{cof}_{31}(\mathbf{A}) + \cdots + a_{nk}\,\mathrm{cof}_{n1}(\mathbf{A}) = \sum_{i=1}^{n} a_{ik}\,\mathrm{cof}_{i1}(\mathbf{A}) = 0,$$

where we find that the $k$-th column appears twice (in the first and $k$-th positions), and the determinant is therefore equal to zero.

It follows that $[\mathbf{Adj}(\mathbf{A})] \cdot \mathbf{A} = |\mathbf{A}| \cdot \mathbf{I}$; and rearranging gives us:

$$\left[\frac{\mathbf{Adj}(\mathbf{A})}{|\mathbf{A}|}\right] \cdot \mathbf{A} = \mathbf{I};$$

where the first matrix (in brackets) is the inverse of $\mathbf{A}$.

The lecture ends here

## *Questions of the Lecture 15*

(L-15) QUESTION 1. Complete the proofs of the exercises of this lecture.

(L-15) QUESTION 2. Consider $\mathbf{A} = \begin{bmatrix}\mathbf{A}_{|1}; & \mathbf{A}_{|2}; & \mathbf{A}_{|3};\end{bmatrix}$ with $\det \mathbf{A} = 2$.

(a) What are $\det(2\mathbf{A})$ and $\det \mathbf{A}^{-1}$?
(b) What is $\det \begin{bmatrix}(3\mathbf{A}_{|1} + 2\mathbf{A}_{|2}); & \mathbf{A}_{|3}; & \mathbf{A}_{|2};\end{bmatrix}$

(L-15) QUESTION 3. The determinant of an $n \times n$ matrix $\mathbf{A}$ is 12 (where $n$ is an even multiple). What is the determinant of $-\mathbf{A}^{\intercal}$? (Justify your answer).
(*MIT Course 18.06 Quiz 2, Fall, 2008*)

(L-15) QUESTION 4. Let $\mathbf{A}$ be a square matrix. Justify whether the following statement is true or false (*Responses that are properly justified will earn points; answers that only indicate the truth value will receive no points*).
$|\mathbf{A}\mathbf{A}^{\intercal}| = |\mathbf{A}|^2$.

(L-15) QUESTION 5. We have a $3 \times 3$ matrix $\mathbf{A} = \begin{bmatrix} a & 1 & 2 \\ b & 3 & 4 \\ c & 5 & 6 \end{bmatrix}$ with $\det \mathbf{A} = 3$. Compute the determinant of the following matrices:

(a) (0.5 pts) $\begin{bmatrix} a-2 & 1 & 2 \\ b-4 & 3 & 4 \\ c-6 & 5 & 6 \end{bmatrix}$
(b) (0.5 pts) $\begin{bmatrix} 7a & 7 & 14 \\ b & 3 & 4 \\ c & 5 & 6 \end{bmatrix}$
(c) (1 pts) $(2\mathbf{A})^{-1}\mathbf{A}^{\intercal}$
(d) (0.5 pts) $\begin{bmatrix} a-2 & 1 & 2 \\ b & 3 & 4 \\ c & 5 & 6 \end{bmatrix}$

(L-15) QUESTION 6.

(a) Find an echelon form of the matrix $\mathbf{A} = \begin{bmatrix} 2 & 3 & 1 \\ 4 & 5 & 2 \\ 4 & 6 & 0 \end{bmatrix}$.
(b) Is $\mathbf{A}$ invertible?
(c) En caso afirmativo calcule $|\mathbf{A}^{-1}|$; en caso contrario calcule $|\mathbf{A}|$
(d) The matrix $\mathbf{C}$ is equal to the product of $\mathbf{A}$ with the *transpose* of matrix $\mathbf{B}$, that is

$$\mathbf{C} = \mathbf{A}\mathbf{B}^{\intercal} \qquad \text{donde} \qquad \mathbf{B} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 0 & 2 & 3 \end{bmatrix}$$

What is the value of the determinant of $\mathbf{C}$? Is $\mathbf{C}$ invertible?

(L-15) QUESTION 7. Compute the determinant of the following matrices using Laplace expansions.

(a) $\begin{bmatrix} 1 & 2 \\ -4 & 3 \end{bmatrix}$
(b) $\begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & 2 \\ 2 & 1 & -2 \end{bmatrix}$
(c) $\begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 2 \\ 2 & 0 & 1 & -2 \end{bmatrix}$

(L-15) Question 8. Compute the following determinant using Laplace expansions:

$$\begin{vmatrix} 0 & 0 & 0 & 3 & 0 \\ -2 & 0 & 0 & 2 & 0 \\ 8 & -1 & 0 & -7 & 2 \\ -1 & 2 & 2 & 3 & 2 \\ 2 & 2 & 3 & 6 & 4 \end{vmatrix}$$

(L-15) Question 9. Compute $\det \mathbf{A} = \begin{vmatrix} 2 & 2 & 0 & 0 \\ 5 & 5 & 0 & 0 \\ 3 & 0 & 1 & 0 \\ 5 & 0 & 0 & 1 \end{vmatrix}$

(L-15) Question 10. Compute the value of $\det \mathbf{A}$ using Laplace expansion

$$\mathbf{A} = \begin{bmatrix} 1 & 1 & 1 & \cdots & 1 \\ 0 & 2 & 2 & \cdots & 2 \\ 0 & 0 & 3 & \cdots & 3 \\ \vdots & & & \ddots & \\ 0 & 0 & 0 & \cdots & n \end{bmatrix}$$

(L-15) Question 11. Consider a $n$ by $n$ matrix $\mathbf{A}_n$ full of 3s in its diagonal, and twos just below the diagonal, and another 2 at the position $(1, n)$; for example, for $n = 4$:

$$\mathbf{A}_4 = \begin{bmatrix} 3 & 0 & 0 & 2 \\ 2 & 3 & 0 & 0 \\ 0 & 2 & 3 & 0 \\ 0 & 0 & 2 & 3 \end{bmatrix}.$$

(a) Find, using the cofactors of the first row, the determinant of $\mathbf{A}_4$.
(b) Find the determinant of $\mathbf{A}_n$ for $n > 4$.

(L-15) Question 12. Consider the following block matrix

$$\mathbf{A} = \begin{bmatrix} \mathbf{B} & \mathbf{0} \\ \mathbf{0} & \mathbf{C} \end{bmatrix}$$

Prove $|\mathbf{A}| = |\mathbf{B}||\mathbf{C}|$.

*Hint.* $\begin{bmatrix} \mathbf{B} & \mathbf{0} \\ \mathbf{0} & \mathbf{C} \end{bmatrix} = \begin{bmatrix} \mathbf{B} & \mathbf{0} \\ \mathbf{0} & \mathbf{I} \end{bmatrix} \begin{bmatrix} \mathbf{I} & \mathbf{0} \\ \mathbf{0} & \mathbf{C} \end{bmatrix}$

(L-15) Question 13. Solve the following linear systems using Cramer's Rule

(a) $\begin{cases} 2x + 5y = 1 \\ x + 4y = 2 \end{cases}.$

(b) $\begin{cases} 2x + y \phantom{+ z} = 1 \\ x + 2y + z = 0 \\ \phantom{x +} y + 2z = 0 \end{cases}.$

(exercise 13 from section 4.4 of Strang (2006))

(L-15) Question 14. Find the inverse of the following matrices using the *adjoint matrix*

(a) $\mathbf{A} = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 3 & 0 \\ 0 & 4 & 1 \end{bmatrix}$

(b) $\mathbf{B} = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$

(exercise 18 from section 4.4 of Strang (2006))

(L-15) QUESTION 15. Consider the matrices $\mathbf{A} = \begin{bmatrix} 1 & 4 & 2 & 3 \\ 2 & 3 & 3 & 7 \\ 0 & 1 & 0 & 3 \\ 0 & 2 & 0 & a \end{bmatrix}$, $\mathbf{B} = \begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{bmatrix}$; and the vector $\boldsymbol{b} = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}$.

(a) ($0.5^{\text{pts}}$) For wich values of $a$ the matrix $\mathbf{A}$ is invertible?
(b) ($1^{\text{pts}}$) Consider $a = 5$. Using the Cramer's rule, compute the fourth coordinate $x_4$ of $\boldsymbol{x}$ for linear system $\mathbf{A}\boldsymbol{x} = \boldsymbol{b}$.
(c) ($1^{\text{pts}}$) Compute $\mathbf{B}^{-1}$. Use the matrix $\mathbf{B}^{-1}$ to solve $\mathbf{B}\boldsymbol{x} = \boldsymbol{b}$.

***End of Questions of the Lecture 15***

# References

Strang, G. (2006). *Linear algebra and its applications.* Thomson Learning, Inc., fourth ed. ISBN 0-03-010567-6.

# Solutions

**Exercise 1.** The zero vector $\mathbf{0}$ in $\mathbb{R}^n$ is a multiple of any other vector $\boldsymbol{x}$ in $\mathbb{R}^n$ since $\mathbf{0} = 0 \cdot \boldsymbol{x}$; it follows, by Property P-3, that $\det \mathbf{A} = 0$:

$$\det\left[\mathbf{A}_{|1} \ldots ; \mathbf{0}; \ldots ; \mathbf{A}_{|n}\right] = \det\left[\mathbf{A}_{|1} \ldots ; 0\boldsymbol{x}; \ldots ; \mathbf{A}_{|n}\right] = 0 \cdot \det\left[\mathbf{A}_{|1} \ldots ; \boldsymbol{x}; \ldots ; \mathbf{A}_{|n}\right] = 0$$

□

**Exercise 2(a)** Since $\mathbf{A}_{\boldsymbol{\tau}_1 \cdots \boldsymbol{\tau}_k} = \mathbf{A}(\mathbf{I}_{\boldsymbol{\tau}_1 \cdots \boldsymbol{\tau}_k}) = \mathbf{A}(\mathbf{I}_{\boldsymbol{\tau}_1}) \cdots (\mathbf{I}_{\boldsymbol{\tau}_k})$, applying repeatedly (1) we get

$$\begin{aligned}
\det(\mathbf{A}_{\boldsymbol{\tau}_1 \cdots \boldsymbol{\tau}_k}) &= \left|\mathbf{A}(\mathbf{I}_{\boldsymbol{\tau}_1}) \cdots (\mathbf{I}_{\boldsymbol{\tau}_k})\right| \\
&= \left|\mathbf{A}(\mathbf{I}_{\boldsymbol{\tau}_1}) \cdots (\mathbf{I}_{\boldsymbol{\tau}_{(k-1)}})\right| \cdot |\mathbf{I}_{\boldsymbol{\tau}_k}| \\
&= \left|\mathbf{A}(\mathbf{I}_{\boldsymbol{\tau}_1}) \cdots (\mathbf{I}_{\boldsymbol{\tau}_{(k-2)}})\right| \cdot |\mathbf{I}_{\boldsymbol{\tau}_{(k-1)}}| \cdot |\mathbf{I}_{\boldsymbol{\tau}_k}| \\
&\quad \vdots \\
&= |\mathbf{A}| \cdot |\mathbf{I}_{\boldsymbol{\tau}_1}| \cdots |\mathbf{I}_{\boldsymbol{\tau}_k}|.
\end{aligned}$$

□

**Exercise 2(b)**

$$|\mathbf{B}| = \det(\mathbf{I}_{\boldsymbol{\tau}_1 \cdots \boldsymbol{\tau}_k}) = |\mathbf{I}_{\boldsymbol{\tau}_1}| \cdots |\mathbf{I}_{\boldsymbol{\tau}_k}|.$$

and, since the determinants of elementary matrices are non-zero, necessarily $|\mathbf{B}| \neq 0$.

□

**Exercise 2(c)** If $\mathbf{B}$ is full rank then it is the product of $k$ elementary matrices $\mathbf{B} = \mathbf{I}_{\boldsymbol{\tau}_1 \cdots \boldsymbol{\tau}_k}$. Hence

$$\det(\mathbf{AB}) = \det(\mathbf{A}\mathbf{I}_{\boldsymbol{\tau}_1 \cdots \boldsymbol{\tau}_k}) = \det(\mathbf{A}_{\boldsymbol{\tau}_1 \cdots \boldsymbol{\tau}_k}) = |\mathbf{A}| \cdot \left(|\mathbf{I}_{\boldsymbol{\tau}_1}| \cdots |\mathbf{I}_{\boldsymbol{\tau}_k}|\right) = |\mathbf{A}| \cdot |\mathbf{B}|.$$

□

**Exercise 3(a)** Since they are both elementary matrices of the same type, $\det(\mathbf{I}_{\boldsymbol{\tau}}) = \det({}_{\boldsymbol{\tau}}\mathbf{I})$.

□

**Exercise 3(b)** Since $\mathbf{B} = \mathbf{I}_{\boldsymbol{\tau}_1 \cdots \boldsymbol{\tau}_k} = (\mathbf{I}_{\boldsymbol{\tau}_1}) \cdots (\mathbf{I}_{\boldsymbol{\tau}_k})$, its determinant is the product of the determinants $\det(\mathbf{I}_{\boldsymbol{\tau}_i})$

$$|\mathbf{B}| = \det\left(\mathbf{I}_{\boldsymbol{\tau}_1}\right) \cdots \det\left(\mathbf{I}_{\boldsymbol{\tau}_k}\right) = \prod_{i=1}^{k} \det\left(\mathbf{I}_{\boldsymbol{\tau}_i}\right).$$

But we also know that $\mathbf{B}^{\mathsf{T}} = {}_{\boldsymbol{\tau}_k \cdots \boldsymbol{\tau}_1}\mathbf{I} = \left({}_{\boldsymbol{\tau}_k}\mathbf{I}\right) \cdots \left({}_{\boldsymbol{\tau}_1}\mathbf{I}\right)$, and thus its determinant is

$$|\mathbf{B}^{\mathsf{T}}| \;=\; \prod_{i=1}^{k} \det\left({}_{\boldsymbol{\tau}_i}\mathbf{I}\right) \;=\; \prod_{i=1}^{k} \det\left(\mathbf{I}_{\boldsymbol{\tau}_i}\right) \;=\; |\mathbf{B}|.$$

□

**(L-14) Question 2.** Since $\mathbf{I} = \mathbf{A}\mathbf{A}^{-1}$, we know that

$$1 = |\mathbf{I}| = |\mathbf{A}||\mathbf{A}^{-1}|;$$

and therefore $\det(\mathbf{A}^{-1}) = \frac{1}{\det(\mathbf{A})}$.

□

**(L-14) Question 3(a)** $\left|\mathbf{A}(\mathbf{B})^2\right| = 2 \cdot (-2)^2 = 8$.
$\left|(\mathbf{AB})^{-1}\right| = \left(|\mathbf{AB}|\right)^{-1} = \frac{1}{-4}$.

□

**(L-14) Question 3(b)** There is not enough information to compute the rank of $\mathbf{A}+\mathbf{B}$. When I say that there is not enough information, I mean we can find two examples of pairs of matrices $(\mathbf{A}_1,\mathbf{B}_1)$ and $(\mathbf{A}_2,\mathbf{B}_2)$ that satisfy the hypotheses: $\det \mathbf{A}_1 = 2 = \det \mathbf{B}_1$ and $\det \mathbf{A}_2 = 2 = \det \mathbf{B}_2$; but $\operatorname{rg}\left(\mathbf{A}_1+\mathbf{B}_1\right) \neq \operatorname{rg}\left(\mathbf{A}_2+\mathbf{B}_2\right)$.

On the other hand, since $|\mathbf{AB}| = -4 \neq 0$, we know $\underset{3\times 3}{\mathbf{AB}}$ is a full rank matrix; hence, its rank is 3.

□

**(L-14) Question 4(a)**

$$\left[\mathbf{A}_1\right] = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow{\overset{\tau}{[(-1)\mathbf{2}+\mathbf{3}]}} \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow{\overset{\tau}{[(-1)\mathbf{2}+\mathbf{1}]}} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \left[\mathbf{I}\right] \quad \Rightarrow \quad \det \mathbf{A}_1 = -1$$

```
A =  Matrix([ [1,0,0], [1,1,1], [0,0,1] ])
Determinante(A)      # esta es una opción
A.determinante()     # esta es otra opción
```

□

**(L-14) Question 4(b)**

$$\begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix} \xrightarrow{\overset{\tau}{\substack{[(2)\mathbf{2}] \\ [(1)\mathbf{1}+\mathbf{2}]}}} \begin{bmatrix} 2 & 0 & 0 \\ -1 & 3 & -1 \\ 0 & -2 & 2 \end{bmatrix} \xrightarrow{\overset{\tau}{\substack{[(3)\mathbf{3}] \\ [(1)\mathbf{2}+\mathbf{3}]}}} \begin{bmatrix} 2 & 0 & 0 \\ -1 & 3 & 0 \\ 0 & -2 & 4 \end{bmatrix} \xrightarrow{\overset{\tau}{\substack{[(3)\mathbf{1}] \\ [(1)\mathbf{2}+\mathbf{1}]}}} \begin{bmatrix} 6 & 0 & 0 \\ 0 & 3 & 0 \\ -2 & -2 & 4 \end{bmatrix}$$

$$\xrightarrow{\overset{\tau}{\substack{[(2)\mathbf{1}] \\ [(1)\mathbf{3}+\mathbf{1}] \\ [(2)\mathbf{2}] \\ [(1)\mathbf{3}+\mathbf{2}]}}} \begin{bmatrix} 12 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 4 \end{bmatrix} \xrightarrow{\overset{\tau}{\substack{[(\frac{1}{12})\mathbf{1}] \\ [(\frac{1}{6})\mathbf{2}] \\ [(\frac{1}{4})\mathbf{3}]}}} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Therefore $\det \mathbf{A}_2 = \frac{1}{2}\cdot\frac{1}{3}\cdot\frac{1}{3}\cdot\frac{1}{2}\cdot\frac{1}{2}\cdot 12\cdot 6\cdot 4 = 4$

□

**(L-14) Question 4(c)**

$$\mathbf{A}_3 = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \xrightarrow{\overset{\tau}{[(-1)\mathbf{2}+\mathbf{3}]}} \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix} \xrightarrow{\overset{\tau}{[(-1)\mathbf{1}+\mathbf{2}]}} \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \xrightarrow{\overset{\tau}{[\mathbf{1}\rightleftharpoons\mathbf{3}]}} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \left[\mathbf{I}\right] \quad \Rightarrow \quad \det \mathbf{A}_3 = -1$$

□

**(L-14) Question 5.** Since all the applied transformations are *Type I*, we have $\mathbf{AE} = \mathbf{I} \Rightarrow \det(\mathbf{A})\cdot 1 = 1$.

□

**(L-14) Question 6(a)** The first one is an elementary matrix, whose determinant is 1.

The second one is a permutation matrix that exchanges two vectors, so its determinant is $-1$.

□

**(L-14) Question 6(b)** $$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ a & b & c & d \end{bmatrix} \xrightarrow{\overset{\tau}{[(\frac{1}{d})\mathbf{4}]}} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ a & b & c & 1 \end{bmatrix} \xrightarrow{\overset{\tau}{\substack{[(-c)\mathbf{4}+\mathbf{3}] \\ [(-b)\mathbf{4}+\mathbf{2}] \\ [(-a)\mathbf{4}+\mathbf{1}]}}} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$ Therefore, the determinant is $d$.

□

**Exercise 4(a)** Como la matriz de orden $n$ es de rango completo, los $n$ elementos de la diagonal principal son pivotes (i.e., distintos de cero).

$$\mathbf{L} = \begin{bmatrix} *_1 & & & \\ \vdots & *_2 & & \\ & & \ddots & \\ \vdots & \vdots & & \\ \vdots & \vdots & \vdots & *_n \end{bmatrix} \qquad \text{donde } *_j \text{ son números distintos de cero.}$$

Dividiendo cada columna $j$-ésima por su pivote $*_j$ para normalizar los pivotes (y compensando dichas transformaciones multiplicando la última fila por cada pivote); y aplicando, en una segunda fase, la eliminación de izquierda a derecha con transformaciones de Tipo I para anular todo lo que queda a la izquierda de los pivotes (ahora basta multiplicar la última fila por 1), llegamos a:

$$\left[\begin{array}{cccc|c} *_1 & & & & \\ \vdots & *_2 & & & \\ \vdots & \vdots & \ddots & & \\ \vdots & \vdots & \vdots & *_n & \\ \hline & & & & 1 \end{array}\right] \xrightarrow[\substack{[(*_1)(\mathbf{n+1})] \\ \vdots \\ [(*_n)(\mathbf{n+1})]}]{\substack{\boldsymbol{\tau} \\ [(\frac{1}{*_1})\mathbf{1}] \\ \vdots \\ [(\frac{1}{*_n})\mathbf{n}]}} \left[\begin{array}{cccc|c} 1 & & & & \\ \vdots & 1 & & & \\ \vdots & \vdots & \ddots & & \\ \vdots & \vdots & \vdots & 1 & \\ \hline & & & & *_1 \cdot *_2 \cdots *_n \end{array}\right] \xrightarrow[\substack{\boldsymbol{\tau} \\ [(1)(\mathbf{n+1})]}]{\substack{\boldsymbol{\tau}_1 \ldots \boldsymbol{\tau}_q \\ \text{(de Tipo I)}}} \left[\begin{array}{cccc|c} 1 & & & & \\ & 1 & & & \\ & & \ddots & & \\ & & & 1 & \\ \hline & & & & *_1 \cdot *_2 \cdots *_n \end{array}\right]$$

Por tanto, si la matriz triangular inferior es de rango completo, su determinante es igual al producto de sus pivotes; es decir, al producto de los elementos de la diagonal.

$$\det(\mathbf{L}) = \text{ producto de los elementos de la diagonal}$$

□

**Exercise 4(b)** Una matriz de orden $n$ y triangular solo puede ser de rango completo si los $n$ elementos de la diagonal son distintos de cero. Por tanto, si la matriz triangular es singular, necesariamente tiene algún cero en su diagonal principal. Como su determinante es cero, por ser singular, su determinante es igual al producto de los elementos de la diagonal (donde uno de ellos es cero).

□

**Exercise 4(c)**

$$\det(\mathbf{U}) = \det\left(\mathbf{U}^{\intercal}\right) = \text{ producto de los elementos de la diagonal}$$

por ser $\mathbf{U}^{\intercal}$ triangular inferior.

□

**Exercise 5.** Expanding by the second column we obtain

$$\det \mathbf{A} = -0\begin{vmatrix} 5 & 2 & 4 \\ 3 & 1 & 2 \\ 5 & 2 & 1 \end{vmatrix} + \begin{vmatrix} 2 & 3 & 2 \\ 3 & 1 & 2 \\ 5 & 2 & 1 \end{vmatrix} - 0\begin{vmatrix} 2 & 3 & 2 \\ 5 & 2 & 4 \\ 5 & 2 & 1 \end{vmatrix} + 3\begin{vmatrix} 2 & 3 & 2 \\ 5 & 2 & 4 \\ 3 & 1 & 2 \end{vmatrix} = 0 + 17 - 0 + 3 \times 4 = 29$$

□

**(L-15) Question 2(a)** $\det\left(2\underset{3\times 3}{\mathbf{A}}\right) = 2^3 \cdot \det \mathbf{A} = 16.$ $\det \mathbf{A}^{-1} = \frac{1}{\det \mathbf{A}} = 1/2.$

□

**(L-15) Question 2(b)**

$$\det\left[(3\mathbf{A}_{|1} + 2\mathbf{A}_{|2});\ \ \mathbf{A}_{|3};\ \ \mathbf{A}_{|2};\right] = \det\left[3\mathbf{A}_{|1};\ \ \mathbf{A}_{|3};\ \ \mathbf{A}_{|2};\right] = 3\det\left[\mathbf{A}_{|1};\ \ \mathbf{A}_{|3};\ \ \mathbf{A}_{|2};\right] = -3\det \mathbf{A} = -6.$$

□

**(L-15) Question 3.**

$$\det(-\mathbf{A}^{\intercal}) = \det(-\mathbf{A}) = (-1)^n \det(\mathbf{A}) = \det(\mathbf{A})$$

since $n$ is an even number.

□

**(L-15) Question 4.** True, since

$$|\mathbf{A}\mathbf{A}^{\intercal}| = |\mathbf{A}||\mathbf{A}^{\intercal}| = |\mathbf{A}||\mathbf{A}| = |\mathbf{A}|^2.$$

□

**(L-15) Question 5(a)**

$$\begin{vmatrix} a-2 & 1 & 2 \\ b-4 & 3 & 4 \\ c-6 & 5 & 6 \end{vmatrix} = \begin{vmatrix} a & 1 & 2 \\ b & 3 & 4 \\ c & 5 & 6 \end{vmatrix} = 3$$

□

**(L-15) Question 5(b)**

$$\begin{vmatrix} 7a & 7 & 14 \\ b & 3 & 4 \\ c & 5 & 6 \end{vmatrix} = 7 \begin{vmatrix} a & 1 & 2 \\ b & 3 & 4 \\ c & 5 & 6 \end{vmatrix} = 7 \times 3 = 21$$

□

**(L-15) Question 5(c)**

$$\left|(2\mathbf{A})^{-1}\mathbf{A}^{\mathsf{T}}\right| = \frac{1}{\det(2\mathbf{A})} \det \mathbf{A} = \frac{1}{2^3 \det \mathbf{A}} \det \mathbf{A} = \frac{1}{8}.$$

□

**(L-15) Question 5(d)**

$$\begin{vmatrix} a-2 & 1 & 2 \\ b & 3 & 4 \\ c & 5 & 6 \end{vmatrix} = \begin{vmatrix} a & 1 & 2 \\ b & 3 & 4 \\ c & 5 & 6 \end{vmatrix} - \begin{vmatrix} 2 & 1 & 2 \\ 0 & 3 & 4 \\ 0 & 5 & 6 \end{vmatrix} = 3 + 4 = 7$$

□

**(L-15) Question 6(a)**

$$\left[\mathbf{A}\right] = \begin{bmatrix} 2 & 4 & 4 \\ 3 & 5 & 6 \\ 1 & 2 & 0 \end{bmatrix} \xrightarrow[\substack{[(-2)\mathbf{1}+\mathbf{2}] \\ [(-2)\mathbf{1}+\mathbf{3}]}]{\boldsymbol{\tau}} \begin{bmatrix} 2 & 0 & 0 \\ 3 & -1 & 0 \\ 1 & 0 & -2 \end{bmatrix} = \left[\mathbf{L}\right]$$

□

**(L-15) Question 6(b)** Since $\mathbf{L}$ has three pivots, $\mathbf{A}$ is invertible.

□

**(L-15) Question 6(c)**

$$|\mathbf{A}^{-1}| = \frac{1}{|\mathbf{A}|} = \frac{1}{\text{producto de los pivotes de } \mathbf{L}} = \frac{1}{2(-1)(-2)} = \frac{1}{4}.$$

□

**(L-15) Question 6(d)**

$$|\mathbf{C}| = |\mathbf{A}\mathbf{B}^{\mathsf{T}}| = |\mathbf{A}||\mathbf{B}^{\mathsf{T}}| = 4 \cdot 0 = 0;$$

since $\mathbf{B}$ has two identical rows. Therefore, $\mathbf{C}$ is not invertible.

□

**(L-15) Question 7(a)**

$$\begin{vmatrix} 1 & 2 \\ -4 & 3 \end{vmatrix} = 1 \cdot 3 - (-4) \cdot 2 = 11$$

□

**(L-15) Question 7(b)** Expanding the determinant by the first row, we have

$$\begin{vmatrix} 1 & -1 & 0 \\ 0 & 1 & 2 \\ 2 & 1 & -2 \end{vmatrix} = 1 \cdot \begin{vmatrix} 1 & 2 \\ 1 & -2 \end{vmatrix} - (-1) \begin{vmatrix} 0 & 2 \\ 2 & -2 \end{vmatrix} + 0 = -8$$

□

**(L-15) Question 7(c)** The quickest approach is to expand the determinant by the second column, since all its elements are zero except for one. Thus, we obtain a minor identical to that in the previous exercise:

$$\begin{vmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 2 \\ 2 & 0 & 1 & -2 \end{vmatrix} = -0 + 1 \cdot \begin{vmatrix} 1 & -1 & 0 \\ 0 & 1 & 2 \\ 2 & 1 & -2 \end{vmatrix} - 0 + 0 = -8$$

□

**(L-15) Question 8.**

$$\begin{vmatrix} 0 & 0 & 0 & 3 & 0 \\ -2 & 0 & 0 & 2 & 0 \\ 8 & -1 & 0 & -7 & 2 \\ -1 & 2 & 2 & 3 & 2 \\ 2 & 2 & 3 & 6 & 4 \end{vmatrix} = -3 \begin{vmatrix} -2 & 0 & 0 & 0 \\ 8 & -1 & 0 & 2 \\ -1 & 2 & 2 & 2 \\ 2 & 2 & 3 & 4 \end{vmatrix} = (-3) \cdot (-2) \begin{vmatrix} -1 & 0 & 2 \\ 2 & 2 & 2 \\ 2 & 3 & 4 \end{vmatrix} = (-3) \cdot (-2) \cdot (2) = 12$$

□

**(L-15) Question 9.**

$$\det \mathbf{A} = \begin{vmatrix} 2 & 2 & 0 & 0 \\ 5 & 5 & 0 & 0 \\ 3 & 0 & 1 & 0 \\ 5 & 0 & 0 & 1 \end{vmatrix} = 1 \cdot \begin{vmatrix} 2 & 2 & 0 \\ 5 & 5 & 0 \\ 3 & 0 & 1 \end{vmatrix} = 1 \cdot 1 \cdot \begin{vmatrix} 2 & 2 \\ 5 & 5 \end{vmatrix} = 1 \cdot 1 \cdot 0 = 0.$$

This result could be anticipated at first glance, as it is clear that the first column is a linear combination of the others.

□

**(L-15) Question 10.** Expanding by the first column

$$\begin{vmatrix} 1 & 1 & 1 & \cdots & 1 \\ 0 & 2 & 2 & \cdots & 2 \\ 0 & 0 & 3 & \cdots & 3 \\ \vdots & & & \ddots & \\ 0 & 0 & 0 & \cdots & n \end{vmatrix} = 1 \cdot \begin{vmatrix} 2 & 2 & \cdots & 2 \\ 0 & 3 & \cdots & 3 \\ \vdots & & \ddots & \\ 0 & 0 & \cdots & n \end{vmatrix} - 0 + 0 - \cdots 0$$

expanding by the first column again

$$= 1 \cdot 2 \cdot \begin{vmatrix} 3 & \cdots & 3 \\ \vdots & \ddots & \\ 0 & \cdots & n \end{vmatrix} - 0 + 0 - \cdots 0$$

and again... and again...

$$1 \cdot 2 \cdot 3 \cdots (n-2) \cdot \begin{vmatrix} (n-1) & (n-1) \\ 0 & n \end{vmatrix} = n!.$$

□

**(L-15) Question 11(a)**

$$|\mathbf{A}_4| = \begin{vmatrix} 3 & 0 & 0 & 2 \\ 2 & 3 & 0 & 0 \\ 0 & 2 & 3 & 0 \\ 0 & 0 & 2 & 3 \end{vmatrix} = 3 \begin{vmatrix} 3 & 0 & 0 \\ 2 & 3 & 0 \\ 0 & 2 & 3 \end{vmatrix} - 2 \begin{vmatrix} 2 & 3 & 0 \\ 0 & 2 & 3 \\ 0 & 0 & 2 \end{vmatrix} = 3 \cdot 27 - 2 \cdot 8 = 65$$

□

**(L-15) Question 11(b)** In general $|\mathbf{A}_n| = 3^n + (-1)^{n-1} 2^n$.

□

**(L-15) Question 12.** Expanding by the last column the first matrix, and expanding by the first column the second matrix, and repeating the expansions with the minors (as in the previous exercise), we get

$$\begin{vmatrix} \mathbf{B} & \mathbf{0} \\ \mathbf{0} & \mathbf{I} \end{vmatrix} = |\mathbf{B}|$$

$$\begin{vmatrix} \mathbf{I} & \mathbf{0} \\ \mathbf{0} & \mathbf{C} \end{vmatrix} = |\mathbf{C}|.$$

Therefore

$$\det \mathbf{A} = \begin{bmatrix} \mathbf{B} & \mathbf{0} \\ \mathbf{0} & \mathbf{C} \end{bmatrix} = \begin{vmatrix} \mathbf{B} & \mathbf{0} \\ \mathbf{0} & \mathbf{I} \end{vmatrix} \begin{vmatrix} \mathbf{I} & \mathbf{0} \\ \mathbf{0} & \mathbf{C} \end{vmatrix} = \det \mathbf{B} \det \mathbf{C}.$$

□

**(L-15) Question 13(a)** On the one hand,

$$\mathbf{A} = \begin{bmatrix} 2 & 5 \\ 1 & 4 \end{bmatrix}; \qquad \boldsymbol{b} = \begin{bmatrix} \mathbf{1} \\ \mathbf{2} \end{bmatrix},$$

on the other hand,

$$\det(\mathbf{A}) = 3; \qquad \begin{vmatrix} \mathbf{1} & 5 \\ \mathbf{2} & 4 \end{vmatrix} = -6; \qquad \begin{vmatrix} 2 & \mathbf{1} \\ 1 & \mathbf{2} \end{vmatrix} = 3.$$

Therefore

$$x = \frac{\begin{vmatrix} \mathbf{1} & 5 \\ \mathbf{2} & 4 \end{vmatrix}}{\det(\mathbf{A})} = \frac{-6}{3} = -2; \qquad y = \frac{\begin{vmatrix} 2 & \mathbf{1} \\ 1 & \mathbf{2} \end{vmatrix}}{\det(\mathbf{A})} = \frac{3}{3} = 1.$$

□

**(L-15) Question 13(b)** On the one hand,

$$\mathbf{A} = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{bmatrix}; \qquad \boldsymbol{b} = \begin{bmatrix} \mathbf{1} \\ \mathbf{0} \\ \mathbf{0} \end{bmatrix},$$

on the other hand,

$$\det(\mathbf{A}) = 4; \qquad \begin{vmatrix} \mathbf{1} & 1 & 0 \\ \mathbf{0} & 2 & 1 \\ \mathbf{0} & 1 & 2 \end{vmatrix} = 3; \qquad \begin{vmatrix} 2 & \mathbf{1} & 0 \\ 1 & \mathbf{0} & 1 \\ 0 & \mathbf{0} & 2 \end{vmatrix} = -2; \qquad \begin{vmatrix} 2 & 1 & \mathbf{1} \\ 1 & 2 & \mathbf{0} \\ 0 & 1 & \mathbf{0} \end{vmatrix} = 1.$$

Therefore, $x = \frac{3}{4}$; $\quad y = \frac{-2}{4} = \frac{-1}{2}$; $\quad z = \frac{1}{4}$.

□

**(L-15) Question 14(a)** $\quad \mathbf{Adj}(\mathbf{A}) = \begin{bmatrix} 3 & -2 & 0 \\ -0 & 1 & -0 \\ 0 & -4 & 3 \end{bmatrix}; \qquad \det(\mathbf{A}) = 3;$

$$\mathbf{A}^{-1} = \frac{\mathbf{Adj}(\mathbf{A})}{|\mathbf{A}|} = \frac{1}{3} \begin{bmatrix} 3 & -2 & 0 \\ 0 & 1 & 0 \\ 0 & -4 & 3 \end{bmatrix}.$$

□

**(L-15) Question 14(b)** $\quad \mathbf{Adj}(\mathbf{B}) = \begin{bmatrix} 3 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 3 \end{bmatrix}; \qquad \det(\mathbf{B}) = 4$

$$\mathbf{B}^{-1} = \frac{\mathbf{Adj}(\mathbf{B})}{|\mathbf{B}|} = \frac{1}{4} \begin{bmatrix} 3 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 3 \end{bmatrix}.$$

Note the inverse of a symmetric matrix is also symmetric.

□

**(L-15) Question 15(a)**

$$\begin{bmatrix} 1 & 4 & 2 & 3 \\ 2 & 3 & 3 & 7 \\ 0 & 1 & 0 & 3 \\ 0 & 2 & 0 & a \end{bmatrix} \to \begin{bmatrix} 1 & 4 & 2 & 3 \\ 0 & -5 & -1 & 1 \\ 0 & 1 & 0 & 3 \\ 0 & 2 & 0 & a \end{bmatrix} \to \begin{bmatrix} 1 & 4 & 2 & 3 \\ 0 & 1 & 0 & 3 \\ 0 & -5 & -1 & 1 \\ 0 & 2 & 0 & a \end{bmatrix} \to \begin{bmatrix} 1 & 4 & 2 & 3 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & -1 & 16 \\ 0 & 0 & 0 & a-6 \end{bmatrix}$$

In order to have a full rank matrix, the parameter $a$ must be different from 6.

$\square$

**(L-15) Question 15(b)** On the one hand

$$\det \mathbf{A} = \begin{vmatrix} 1 & 4 & 2 & 3 \\ 2 & 3 & 3 & 7 \\ 0 & 1 & 0 & 3 \\ 0 & 2 & 0 & 5 \end{vmatrix} = 1 \begin{vmatrix} 3 & 3 & 7 \\ 1 & 0 & 3 \\ 2 & 0 & 5 \end{vmatrix} - 2 \begin{vmatrix} 4 & 2 & 3 \\ 1 & 0 & 3 \\ 2 & 0 & 5 \end{vmatrix} = -3 \begin{vmatrix} 1 & 3 \\ 2 & 5 \end{vmatrix} - 2 \cdot (-2) \begin{vmatrix} 1 & 3 \\ 2 & 5 \end{vmatrix} = \begin{vmatrix} 1 & 3 \\ 2 & 5 \end{vmatrix} = -1.$$

On the other hand

$$\begin{vmatrix} 1 & 4 & 2 & 1 \\ 2 & 3 & 3 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 2 & 0 & 0 \end{vmatrix} = -1 \cdot \begin{vmatrix} 2 & 3 & 3 \\ 0 & 1 & 0 \\ 0 & 2 & 0 \end{vmatrix} = -2 \begin{vmatrix} 1 & 0 \\ 2 & 0 \end{vmatrix} = 0$$

Consequently, $x_4 = \frac{0}{-1} = 0$.

$\square$

**(L-15) Question 15(c)**

$$\left[\begin{array}{cccc} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ \hline 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right] \xrightarrow[\substack{[(-1)\mathbf{2}+\mathbf{3}] \\ [(-1)\mathbf{1}+\mathbf{4}]}]{\tau} \left[\begin{array}{cccc} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & -1 & 0 \\ \hline 1 & 0 & 0 & -1 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right] \xrightarrow[\substack{[\mathbf{1} \rightleftharpoons \mathbf{2}] \\ [\mathbf{3} \rightleftharpoons \mathbf{4}]}]{\tau} \left[\begin{array}{cccc} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & -1 \\ \hline 0 & 1 & -1 & 0 \\ 1 & 0 & 0 & -1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{array}\right] \xrightarrow[\substack{[(1)\mathbf{4}+\mathbf{1}] \\ [(-1)\mathbf{4}]}]{\tau} \left[\begin{array}{cccc} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ \hline 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \end{array}\right].$$

Hence $\mathbf{B}^{-1} = \begin{bmatrix} 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$ and thus, multiplying by $\mathbf{B}^{-1}$ we get $\mathbf{B}\boldsymbol{x} = \boldsymbol{b} \Rightarrow \mathbf{B}^{-1}\mathbf{B}\boldsymbol{x} = \mathbf{B}^{-1}\boldsymbol{b} \Rightarrow \boldsymbol{x} = \mathbf{B}^{-1}\boldsymbol{b}$:

$$\boldsymbol{x} = \mathbf{B}^{-1}\boldsymbol{b} = \begin{bmatrix} 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}.$$

$\square$